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RESEARCH ARTICLE

# Multicriteria analysis for flood risk map development: a hierarchical method applied to Brazilian cities

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## Abstract

Floods have caused socio-economic and environmental damage globally and, thus, require research. Several factors influence flooding events, such as extreme rainfall, physical characteristics, and local anthropogenic factors; therefore, such factors are essential for mapping flood risk areas and enabling measures that mitigate the damage they cause. This study aimed to map and analyze regions susceptible to flood risk in three different study areas belonging to the same Atlantic Forest biome, in which flood disasters are recurrent. Due to the presence of numerous factors, a multicriteria analysis using the Analytical Hierarchical Process was conducted. First, a geospatial database was composed of layers of elevation, slope, drainage distance, soil drainage, soil hydrological group, precipitation, relief, and land use and cover. Flood risk maps for the study area were then generated, and patterns in the study areas were verified, with the greatest influence being exerted by intense precipitation on consecutive days, elevation at the edges of the channel with low altimetric variation and a flat combination, densely built areas close to the banks of the main river, and an expressive water mass in the main watercourse. The results demonstrate that these characteristics together can indicate the occurrence of flooding events.

**Keywords** Analytical Hierarchical Process · Disasters · Susceptibility map · Extreme rainfall · Geospatial database · Physical characteristics

## Introduction

Floods are caused by intense rains that cause riverbeds to overflow, thereby damaging the surrounding areas. The uncontrolled growth of cities, combined with high rates of soil sealing, increases the risk of flooding. Owing to the lack of proper management of water resources, these events have become commonplace in several places, demanding studies on this subject. To prevent or minimize disasters, prior knowledge of areas most susceptible to flood risks is essential, as they enable adequate management for controlling these disasters. According to Lima et al. (2022), the occurrence of flood disasters impacts human health and urban infrastructure, resulting in the suspension of public transport, damage to property, and spread of diseases, in addition to the financial costs of insurance and public compensation schemes.

In general, risk management is defined as a process that deals with unforeseen events, allowing for the prediction of future scenarios within an acceptable range of variability. Therefore, if an unfavorable event occurs, actions put in place beforehand can reduce its impacts (Almeida et al. 2015). Almeida et al. (2015) stated that risk control involves planning risk management activities and implementing risk resolution actions, be it eliminating or mitigating them, with constant monitoring of the variables identified during the assessment phase. According to Oliveira et al. (2021), risk analyses should precede the scaling of projects that exhibit a degree of complexity (such as unique works of art) once the risk associated with damage caused by a hydrologic event is higher than that predicted for the project during its lifetime. Oliveira et al. (2021) also highlighted the importance of predicting hydrological quantities of considerable magnitude,

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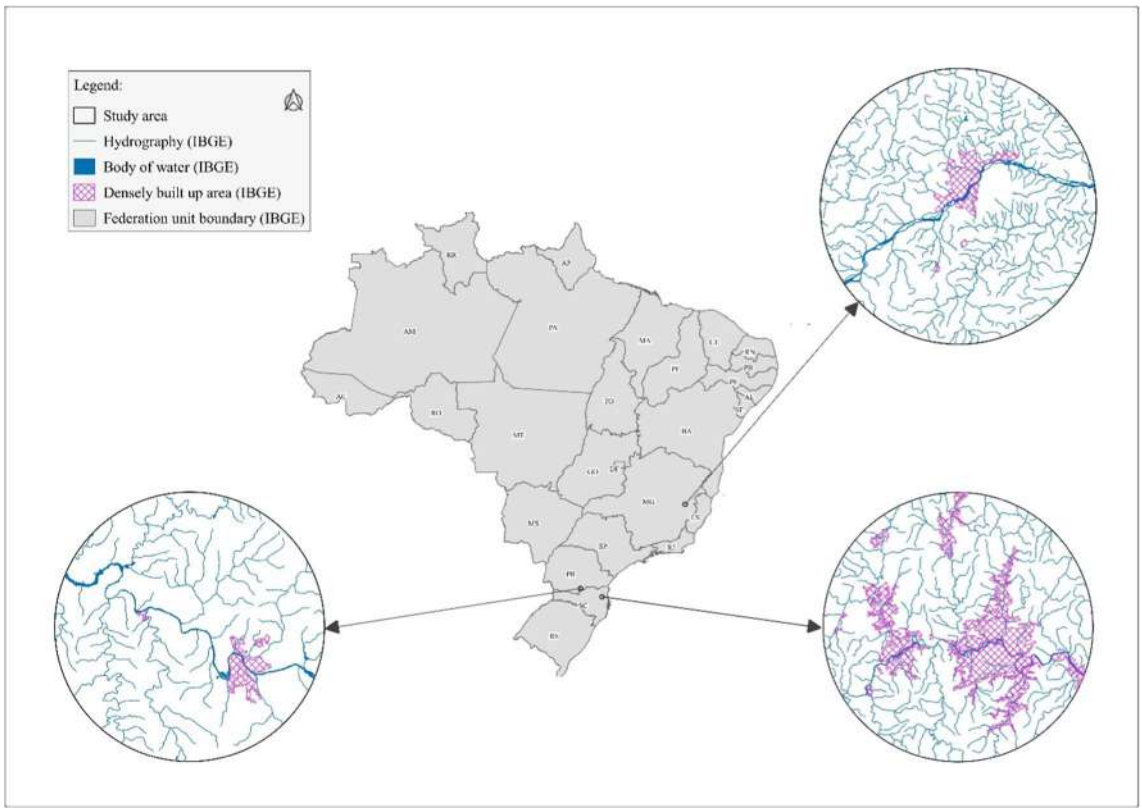



**Fig. 1** Location of the study area: União da Vitória, PR; Blumenau, SC; Governador Valadares, MG

such as maximum daily precipitation, that can cause flooding in a certain region. According to Canabrava Neto (2021), the surface runoff flow component is specifically important during and immediately after rain peaks, when it causes a rapid increase in water flow, characterizing flood peaks in hydrographs.

To determine areas most susceptible to flood risks, mapping of these areas is essential. Therefore, it is necessary to use geoprocessing tools and construct a geospatial database to model the input information, which must be previously selected and screened to determine the constraints of the problem in question and, thus, extract areas most susceptible to these flood events. The joint applications of the geographic information system (GIS) and Multi-Criteria Decision Support system (MCDSS) provide support to researchers in solving complex problems with large amounts of data (Mahmoud and Gan 2018). Remote sensing and GIS techniques are useful for surveying areas susceptible to flooding and for generating flood susceptibility maps (Harley and Samanta 2018).

Almeida et al. (2022) reviewed times of concentration and verified that many studies on floods, flash floods, and flood vulnerability maps have been developed in recent years worldwide. According to Almeida et al. (2021), a physical characterization of a watershed may provide important data for preventing and assessing environmental risks, and analyses of the hydrological and physiographic parameters of a watershed yield a large number of variables. Almeida et al. (2017) mentioned that flow catchment behavior depends on the intensity and distribution of rainfall, both in space and time, and on the physical features of the catchment, including geology, soils, vegetation, and climate. Studies on water surface quantity, quality, and behavior should consider many

**Table 1** SRTM images and altimetry dimensions

| Study area | União da Vitória, PR | Blumenau, SC | Governador Valadares, MG |
|---|---|---|---|
| Minimum height (m) | 710 | 0 | 131 |
| Maximum height (m) | 1270 | 890 | 1109 |
| Amplitude | 560 | 890 | 978 |
| SRTM images | s26 w052 1 arc, s27 w052 1arc, s27 w051 1 arc | s27 w049 1arc, s27 w050 1arc, s28 w050 1 arc | s19 w042 1 arc, s19 w043 1 arc, s20 w042 1 arc, s20 w043 1 arc |

variables, and it is essential to identify relationships between the variables (Costa et al. 2022; Almeida et al. 2021) and to verify the relevance and importance of each variable in a study area (Lima et al. 2022).

Flood risk mapping involves numerous factors that require cross-information. This information lacks geographic representation, indicating the usefulness of GIS and multicriteria analyses. Several research groups have made advances in systematic watershed analyses using GIS software for morphometric analyses to verify regions susceptible to flooding, allowing for the prioritization of basins according to their flood potential (Almeida et al. 2022). A flood risk assessment aims to identify areas or locations that are likely to flood in the future (Chakraborty and Mukhopadhyay 2019). Surface runoff, flow accumulation, soil type, altitude, distance from the drainage network, drainage density, land use, slope, and geology are the main contributing factors influencing the development of flood susceptibility maps (Mahmoud and Gan, 2018). Dou et al. (2018) used seven parameters that contributed to the occurrence of floods and identified the layers of rainfall intensity, flow accumulation, and distance from the river network as being the main determining factors. According to Mahmoud and Gan (2018), surface runoff and accumulated flow tend to have greater weights than do other factors, such as soil type, altitude, distance from the drainage network, and drainage density. It is recommended to consider six to ten factors when identifying areas susceptible to flooding so that no susceptibility factor is an erroneous determinant (Mahmoud and Gan 2018). Saaty (1977) identified two advantages in grouping complex problems in a hierarchical way: the efficiency of a paired comparison and the greater consistency that depends on the premise of the mind's ability to compare $7 \pm 2$ criteria simultaneously.

Flood disaster and mitigation are complex and interdisciplinary, involving several conflicting factors, in addition to the several steps in the selection process, making multicriteria analyses suitable for modeling such problems (Lima et al. 2022). According to Lopes et al. (2022), multicriteria analyses are employed in this type of problem because they allow for a more objective discussion, represent the main characteristics of the analyzed system, and numerically translate the options of the main decision makers in relation to the parameters that describe the main characteristics of the studied object or project.

In a survey of areas at risk of flooding, numerous conditioning factors are considered and are usually selected according to the researcher's expertise; however, it should be noted that information on altitude, slope, and land use is commonly incorporated into the input data (Dou et al. 2018; Mahmoud and Gan 2018; Arabameri et al. 2019; Surwase et al. 2019; Mohamed and EL-Raey 2020). Vojtek and Vojteková (2019) stated that the basic factors relevant to mapping areas susceptible to floods are slope, elevation, land use and cover, distance from rivers or river network density, and lithology. The authors also stated that the number of flood-conditioning factors and which factors should be employed is debatable, and they claim that there is no standard for choosing which factors should be used in the analysis of flood susceptibility.

Flood risk maps both present areas susceptible to flooding and provide an opportunity to verify the functioning of deployed flood defense systems (Chakraborty and Mukhopadhyay 2019). In their study of flood risk analysis, Chakraborty and Mukhopadhyay (2019) focused on three basic elements: flood risk, susceptibility and flood hazard, and the information necessary for flood management. The areas were evaluated and mapped to provide a decision-making system for identifying the trajectory of these disasters.

**Table 2** Slope classes

| Slope classes | Percentage limits (%) |
|---|---|
| Plan | 0–3% |
| Smooth wavy | 3–8% |
| Wavy | 8–20% |
| Strong wavy | 20–45% |
| Mountainous | 45–75% |
| Escarpment | > 75% |

Source: (EMBRAPA, 2006)

**Table 3** Soil drainage classes and assigned scores

| Soil drainage classes | Score |
|---|---|
| Excessively drained | 0.5 |
| Somewhat excessively drained | 1.0 |
| Very well drained | 1.5 |
| Well drained | 2.0 |
| Moderately drained | 2.5 |
| Somewhat poorly drained | 3.0 |
| Poorly drained | 3.5 |
| Very poorly drained | 4.0 |

**Table 4** Soil hydrological groups and assigned scores

| Soil hydrological group | Score |
|---|---|
| Group A | 1.0 |
| Group B | 2.0 |
| Group C | 3.0 |
| Group D | 4.0 |
| Urban area | 4.5 |
| Continental bodies of water | 5.0 |

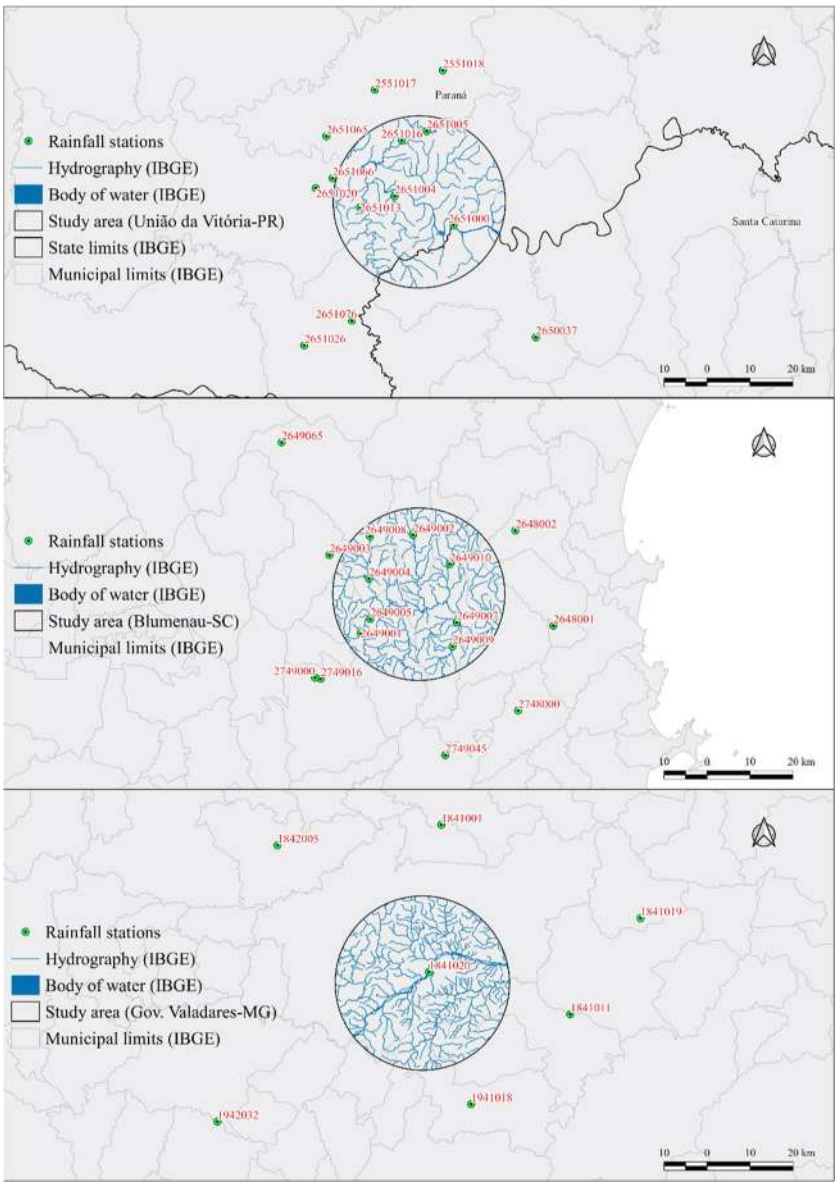



**Fig. 2** Location of rainfall stations in the study areas of União da Vitória, PR; Blumenau, SC; and Governador Valadares, MG

GIS and multicriteria analysis tools are widely used to map flood risk areas and demonstrate consistency in the results. Their applicability was proven in the studies by Dou et al. (2018), Mahmoud and Gan (2018), Arabameri et al. (2019), Surwase et al. (2019), Mohamed and EL-Raey (2020), and Msabi and Makonyo (2021). In this way, they can be subsidies both for mitigation plans for damage caused by flooding events and for verifying the efficiency of a prevention system already in place at a site under analysis.

Msabi and Makonyo (2021) considered seven influencing factors for mapping the flood susceptible areas in the Dodoma region, located in the central part of Tanzania. Hammami et al. (2019) developed a spatial multi-criteria decision production prototype, considering eight criterious, for the evaluation of flooding susceptibility for the city of Tunis, North-East of Tunisia, North Africa. Vojtek and Voiteková (2019) chose seven flood conditioning factors to define the flood susceptibility zones for the territory of

**Table 5** Random Consistency Index

| *n* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Random Consistency Index (R.I) | 0.00 | 0.00 | 0.58 | 0.90 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 | 1.49 |

Fonte: Saaty (1991)

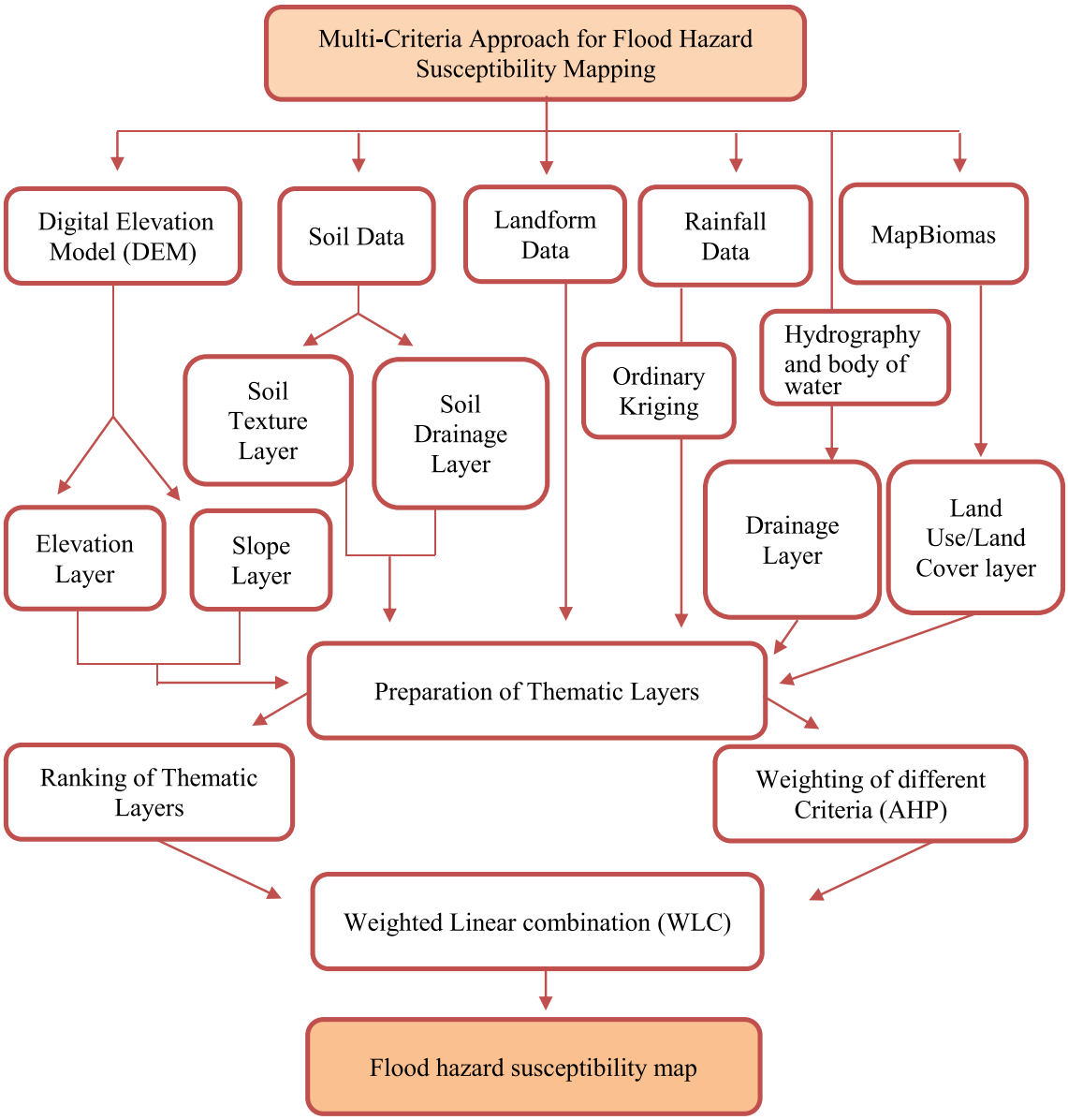



**Fig. 3** Methodological flowchart for mapping flood susceptibility using MCA and WLC

**Table 6** AHP value scale for a paired comparison

| Intensity of importance | Definition | Explanation |
|---|---|---|
| 1 | Equal importance | Both activities contribute equally to the objective |
| 3 | Moderate importance | Experience and judgment slightly favor one activity over another |
| 5 | Essential importance | Experience and judgment strongly favor one activity over another |
| 7 | Very strong importance | An activity is strongly favored and its dominance is demonstrated in practice |
| 9 | Extreme importance | Evidence favoring one activity over another is of the highest possible claim order. |
| 2, 4, 6, 8 | Intermediate values between judgments | When a compromise is required |

Fonte: Adapted from Saaty (1990)

**Table 7** Completed AHP worksheet

| | Altitude | Slope | Soil drainage | Hydrological group | Drainage distance | Precipitation | Relief | Land use and occupation |
|---|---|---|---|---|---|---|---|---|
| Altitude | 1 | 1/5 | 1/5 | 1/6 | 1/7 | 1/9 | 1/5 | 1/7 |
| Slope | 5 | 1 | 1/5 | 1/5 | 1/7 | 1/9 | 1/5 | 1/7 |
| Soil drainage | 5 | 5 | 1 | 2 | 1/7 | 1/9 | 3 | 1/7 |
| Hydrological group | 6 | 5 | ½ | 1 | 1/7 | 1/9 | 3 | 1/7 |
| Drainage distance | 7 | 7 | 7 | 7 | 1 | 1/5 | 7 | 5 |
| Precipitation | 9 | 9 | 9 | 9 | 5 | 1 | 9 | 7 |
| Relief | 5 | 5 | 1/3 | 1/3 | 1/7 | 1/9 | 1 | 1/5 |
| Land use and occupation | 7 | 7 | 7 | 7 | 1/5 | 1/7 | 5 | 1 |

Slovakia using a multi-criteria approach, particularly the Analytical Hierarchy Process (AHP) technique, and geographic information systems (GIS). In the study of Swain et al. (2020), five main flood-causing criteria, with 21 subcriteria, were broadly selected to investigate a flood susceptible region in Bihar, India. Souissi et al. (2020) presented a study that prepares a flood hazard susceptibility map of the Gabes region, located in the southeast part of Tunisia, using eight factors in the flood modeling.

The variables that control the hydrological behavior of a region are many and often interdependent. In regions with a tropical climate, there is the seasonal occurrence of heavy rains, accompanied by floods and floods that cause material and non-material damage to the population. Brazil is a country of continental territorial extension, presenting specific characteristics for each region, such as precipitation volume, geological and pedological formation, relief, population density, characteristics of urban buildings, availability of data series, and monitoring network for analysis. Thus, it is necessary for public managers to have tools that help in urban planning, as well as in the implementation of mitigation measures in areas with potential risk of disaster occurrence. Vegetation influences the hydrological regime of a region, controlling floods, maintaining flows during dry periods, and providing good quality water and total precipitation. Thus, the biome was considered in this study as a limiting characteristic for the selection of regions with flood records. The objective of this study was to analyze and map the risk of flooding in three areas belonging to the same Atlantic Forest biome: União da Vitória, PR; Blumenau, SC; and Governador Valadares, MG. These areas are delimited from a circumference with a radius of 20 km, which encompassed the urbanized areas of these cities. Such areas have recurring flood events, as described in the Disaster Information Form (FIDE) (Brasil 2020; FIDE 2013, 2014). To map the risk areas, a multicriteria analysis, specifically the Analytical Hierarchical Process (AHP), was applied. In this study, eight main criteria groups were considered. First, altitude, slope, precipitation, soil drainage, soil hydrological group, drainage distance, relief, and land use and cover were considered to identify the areas most susceptible to flooding. From the mapping, it was possible to analyze similarities among the input data, determine which criteria were most important and common in the three areas, and verify the results obtained from mapping the areas with high and very high flood risks.

## Methodology

### Study area

Three study areas with recurring floods were selected, and occurrence records from FIDE confirmed that these areas belong to the same Atlantic Forest biome.

The selected areas comprise the urbanized municipal portion of União da Vitória, PR; Blumenau, SC; and Governador Valadares, MG, where the occurrence of floods and their consequences are so recurrent that they are considered natural hydrological disasters. However, these areas lack research, control, and preventive measures. To delimit

**Table 8** Paired comparison results

| Class | AHP |
|---|---|
| Altitude | 0.0173 |
| Slope | 0.0311 |
| Soil drainage | 0.0492 |
| Hydrological group | 0.0642 |
| Drainage distance | 0.0685 |
| Precipitation | 0.1533 |
| Relief | 0.2167 |
| Land use and occupation | 0.3998 |

**Table 9** Results of altitude classes

| Altitude classes (m) | Scores | Study area | km$^2$ | % |
|---|---|---|---|---|
| 700–800 | 5 | União da Vitória, PR | 329.2 | 26.18% |
| 800–900 | 4 | União da Vitória, PR | 373.9 | 29.73% |
| 900–1000 | 3 | União da Vitória, PR | 374.6 | 29.79% |
| 1000–1100 | 2 | União da Vitória, PR | 158.2 | 12.58% |
| 1100–1200 | 1 | União da Vitória, PR | 21.7 | 1.73% |
| 0–200 | 5 | Blumenau, SC | 765.7 | 60.94% |
| 200–400 | 4 | Blumenau, SC | 380.7 | 30.30% |
| 400–600 | 3 | Blumenau, SC | 99.4 | 7.91% |
| 600–800 | 2 | Blumenau, SC | 10.6 | 0.84% |
| 800–1000 | 1 | Blumenau, SC | 0.2 | 0.02% |
| 100–300 | 5 | Governador Valadares, MG | 1068.9 | 83.15% |
| 300–500 | 4 | Governador Valadares, MG | 194.5 | 15.13% |
| 500–700 | 3 | Governador Valadares, MG | 12.8 | 1.00% |
| 700–900 | 2 | Governador Valadares, MG | 7.7 | 0.60% |
| 900–1100 | 1 | Governador Valadares, MG | 1.6 | 0.13% |

these three areas, a circle with a diameter of 40 km was created using the Quantum GIS (QGIS 2020) software to fully include the urban areas of the selected cities, encompassing the upstream portion close to the region.

The União da Vitória study area has an area of approximately 1258 km$^2$ and 84% of its area is located in the state of Paraná, while the remaining 16% is in the state of Santa Catarina, which belongs to the Atlantic Forest biome, and has the Iguaçu River as its main watercourse (Fig. 1). Parts of the municipalities of União da Vitória, PR; General Carneiro, PR; Porto Vitória, PR; Paula Freitas, PR; Cruz Machado, SC; Bituruna, PR; and Porto União, SC, were part of this study area. According to CEPED (2013), the municipality most affected by recurrent floods in the period 1991–2012 in the state of Paraná was União da Vitória, with ten occurrences, where floods reached urban and rural areas, on the banks of the Iguaçu River that cuts through its territory. With regard to material damage, in 1992, 4500 buildings were damaged, and in 2010, 551 buildings were damaged and one destroyed due to floods in the municipality.

The Blumenau study area has an area of approximately 1257 km$^2$ and is located in the State of Santa Catarina, which belongs to the Atlantic Forest biome and has the Itajaí-açu River as its main watercourse (Fig. 1). The

**Table 10** Slope (EMBRAPA) in the study areas

| EMBRAPA slope classes | Score | Study area | km$^2$ | % |
|---|---|---|---|---|
| Plan | 5 | União da Vitória, PR | 71.3 | 5.67% |
| Smooth wavy | 4 | União da Vitória, PR | 243.0 | 19.32% |
| Wavy | 3 | União da Vitória, PR | 488.9 | 38.88% |
| Strong wavy | 2 | União da Vitória, PR | 411.0 | 32.68% |
| Mountainous | 1 | União da Vitória, PR | 42.0 | 3.34% |
| Steep | 1 | União da Vitória, PR | 1.5 | 0.12% |
| Plan | 5 | Blumenau, SC | 54.7 | 4.35% |
| Smooth wavy | 4 | Blumenau, SC | 177.8 | 14.15% |
| Wavy | 3 | Blumenau, SC | 315.8 | 25.13% |
| Strong wavy | 2 | Blumenau, SC | 563.3 | 44.83% |
| Mountainous | 1 | Blumenau, SC | 139.3 | 11.09% |
| Steep | 1 | Blumenau, SC | 5.6 | 0.45% |
| Plan | 5 | Governador Valadares, MG | 84.4 | 6.56% |
| Smooth wavy | 4 | Governador Valadares, MG | 172.2 | 13.39% |
| Wavy | 3 | Governador Valadares, MG | 349.5 | 27.18% |
| Strong wavy | 2 | Governador Valadares, MG | 610.2 | 47.47% |
| Mountainous | 1 | Governador Valadares, MG | 66.6 | 5.18% |
| Steep | 1 | Governador Valadares, MG | 2.7 | 0.21% |

**Table 11** Distance from drainage in the study areas

| Drainage distance (m) | Study area | km$^2$ | % |
|---|---|---|---|
| < 125 | União da Vitória, PR | 176.5 | 14.04% |
| 125–250 | União da Vitória, PR | 150.2 | 11.95% |
| 250–375 | União da Vitória, PR | 142.7 | 11.35% |
| 375–500 | União da Vitória, PR | 132.7 | 10.55% |
| > 500 | União da Vitória, PR | 655.4 | 52.11% |
| < 125 | Blumenau, SC | 246.9 | 19.65% |
| 125–250 | Blumenau, SC | 219.3 | 17.45% |
| 250–375 | Blumenau, SC | 197.6 | 15.72% |
| 375–500 | Blumenau, SC | 167.7 | 13.35% |
| > 500 | Blumenau, SC | 425.2 | 33.83% |
| < 125 | Governador Valadares, MG | 288.5 | 22.44% |
| 125–250 | Governador Valadares, MG | 245.7 | 19.11% |
| 250–375 | Governador Valadares, MG | 213.8 | 16.63% |
| 375–500 | Governador Valadares, MG | 171.7 | 13.36% |
| > 500 | Governador Valadares, MG | 365.9 | 28.46% |

Itajaí-açu River is highly vulnerable to flooding. Portions of the municipalities of Blumenau, SC, Ascurra, SC; Benedito Novo, SC; Gaspar, SC; Ilhota, SC; Indaial, SC; Luiz Alves, SC; Massaranduba, SC; Pomerode, SC; Rio dos Cedros, SC; Rodeo, SC; and Timbó, SC were included in this study area. These municipalities are located in the Mesoregion of the State of Santa Catarina called Vale do Itajaí. The State of Santa Catarina has 1696 official records of torrents characterized as a disaster, between the years 1991 and 2012, with 27% of these records in the Mesoregion of Vale do Itajaí, and 16 records of torrents in Blumenau City (CEPED, 2013b). In the same period, 449 official records of floods were recorded, with 23% of them in the Itajaí Valley. In 2008, 5209 people were homeless, 25000 were displaced, 24 died, and 103000 were affected in Blumenau City due to floods. Blumenau also recorded significant human damage due to flood events, with 396 people being homeless and 16828 affected in 2001; the same numbers were recorded in 2005; in 2011, 668 people were homeless and 302000 affected; in 2013, 281 were homeless, 12340 displaced, and 31000 affected.

**Table 12** Soil drainage in the study areas

| Class | km$^2$ | Drainage |
|---|---|---|
| Pedology— União da Vitória, Paraná | | |
| Urban area | 26.1 | - |
| CHa—Cambissolo Húmico Alumínico | 64.7 | Moderately drained |
| Continental water bodies | 19.5 | - |
| CXa—Cambissolo Háplico Alumínico | 249.7 | Somewhat excessively drained |
| CXal—Cambissolo Háplico Alítico | 104.5 | Moderately drained |
| CXbd—Cambissolo Háplico Tb Distrófico | 494.3 | Somewhat excessively drained |
| GXbd—Gleissolo Háplico Tb Distrófico | 65.2 | Very poorly drained |
| NBaf—Nitossolo Bruno Aluminoférrico | 11.3 | Moderately drained |
| NVdf—Nitossolo Vermelho Distroférrico | 73.0 | Well drained |
| RLd—Neossolo Litólico Distrófico | 20.4 | Very well drained |
| RLdh—Neossolo Litólico Distro-Úmbrico | 43.8 | Very well drained |
| RLe—Neossolo Litólico Eutrófico | 85.3 | Overly drained |
| Pedology—Blumenau, Santa Catarina | | |
| Urban area | 138.4 | - |
| Continental water bodies | 9.3 | - |
| CXa—Cambissolo Háplico Alumínico | 705.6 | Somewhat excessively drained |
| GXbd—Gleissolo Háplico Tb Distrófico | 18.3 | Very poorly drained |
| PVAa—Argissolo Vermelho-Amarelo Alumínico | 62.0 | Moderately drained |
| PVAd—Argissolo Vermelho-Amarelo Distrófico | 323.1 | Moderately drained |
| Pedology—Governador Valadares, Minas Gerais | | |
| Urban area | 38.6 | - |
| Continental water bodies | 19.3 | - |
| CXbe—Cambissolo Háplico Tb Eutrófico | 41.9 | Well drained |
| LAa—Latossolo Amarelo Alumínico | 23.3 | Well drained |
| LAd—Latossolo Amarelo Distrófico | 68.9 | Well drained |
| LVAd—Latossolo Vermelho-Amarelo Distrófico | 54.4 | Very well drained |
| PVAd—Argissolo Vermelho-Amarelo Distrófico | 171.1 | Well drained |
| PVe—Argissolo Vermelho Eutrófico | 838.8 | Well drained |
| RYve—Neossolo Flúvico Ta Eutrófico | 25.0 | Somewhat poorly drained |
| TXp—Luvissolo Háplico Pálico | 4.4 | Poorly drained |

The Governador Valadares study area has an area of approximately 1286 km$^2$ and is located in the State of Minas Gerais, which belongs to the Atlantic Forest biome, and has the Rio Doce as its main watercourse (Fig. 1). The Rio Doce River is highly vulnerable to flooding Portions of the municipalities of Governador Valadares, MG; Alpercata, MG; Capitão Andrade, MG; Engenheiro Caldas, MG; Fernandes Tourinho, MG; Itanhomi, MG; and Tumiritinga, MG, were part of this study area. In 2012, Governador Valadares registered 389 homeless, 25671 housed, 74416 affected, and 9254 buildings damaged due to torrents. In 2005, due to the flood, Governador Valadares registered 1052 homeless, 50595 affected, and in 2009, 478 homeless and 43696 were registered (CEPED, 2013c).

## Identification of flood risk areas

To identify flood risk areas and map susceptibility, a step-by-step methodology based on Harley and Samanta (2018) was applied. In the identification and mapping of areas susceptible to flooding, a geospatial database with different information on the area to be mapped and analyzed is necessary, since countless variables are necessary to identify more vulnerable areas, in such a way that these data are processed using GIS. Thus, it is possible to perform map algebra and obtain a flood risk map. Therefore, for the proposed methodology as well as data on altitude, slope, drainage distance, soil drainage, hydrological group, precipitation, relief, land use, and occupation were used, making up a total of eight layers.

**Table 13** Soil hydrological group in the study areas

| Classes | km$^2$ | Hydrological group | Score |
|---|---|---|---|
| Pedology—União da Vitória, Paraná | | | |
| Urban area | 26.1 | - | 4.5 |
| CHa—Cambissolo Húmico Alumínico | 64.7 | C | 3 |
| Continental water bodies | 19.5 | - | 5 |
| CXa—Cambissolo Háplico Alumínico | 249.7 | C | 3 |
| CXal—Cambissolo Háplico Alítico | 104.5 | C | 3 |
| CXbd—Cambissolo Háplico Tb Distrófico | 494.3 | C | 3 |
| GXbd—Gleissolo Háplico Tb Distrófico | 65.2 | D | 4 |
| NBaf—Nitossolo Bruno Aluminoférrico | 11.3 | B | 2 |
| NVdf—Nitossolo Vermelho Distroférrico | 73.0 | B | 2 |
| RLd—Neossolo Litólico Distrófico | 20.4 | D | 4 |
| RLdh—Neossolo Litólico Distro-Úmbrico | 43.8 | D | 4 |
| RLe—Neossolo Litólico Eutrófico | 85.3 | D | 4 |
| Pedology—Blumenau, Santa Catarina | | | |
| Urban area | 138.4 | - | 4.5 |
| Continental water bodies | 9.3 | - | 5 |
| CXa—Cambissolo Háplico Alumínico | 705.6 | C | 3 |
| GXbd—Gleissolo Háplico Tb Distrófico | 18.3 | D | 4 |
| PVAa—Argissolo Vermelho-Amarelo Alumínico | 62.0 | C | 3 |
| PVAd—Argissolo Vermelho-Amarelo Distrófico | 323.1 | C | 3 |
| Pedology—Governador Valadares, Minas Gerais | | | |
| Urban area | 38.6 | - | 4.5 |
| Continental water bodies | 19.3 | - | 5 |
| CXbe—Cambissolo Háplico Tb Eutrófico | 41.9 | D | 4 |
| LAa—Latossolo Amarelo Alumínico | 23.3 | A | 1 |
| LAd—Latossolo Amarelo Distrófico | 68.9 | A | 1 |
| LVAd—Latossolo Vermelho-Amarelo Distrófico | 54.4 | A | 1 |
| PVAd—Argissolo Vermelho-Amarelo Distrófico | 171.1 | C | 3 |
| PVe—Argissolo Vermelho Eutrófico | 838.8 | B | 2 |
| RYve—Neossolo Flúvico Ta Eutrófico | 25.0 | C | 3 |
| TXp—Luvissolo Háplico Pálico | 4.4 | D | 4 |

The original vector layers were converted into the raster format, and 30 m of output spatial resolution was established, considering that the data used in this study, such as the images used to extract altitude and slope and the images used to obtain the land and occupation, both had spatial resolutions of 30 m. Therefore, the developed flood maps had the same spatial resolutions.

The following input layers were used: altitude, slope, drainage distance, soil drainage, hydrological group, precipitation, relief and land use, and occupation. The flood risk maps of the study areas were then generated using the QGIS software from the map algebra tool.

The methodology used in this study adopted the multicriteria approach (MCA) method, which has been successful in studies on analyzing and mapping of flood susceptibility (Vojtek and Vojteková, 2019; Souissi et al. 2020; Das and Gupta 2021; Nsangou et al. 2022) and is an effective tool for managing and monitoring floods.

Based on the knowledge of the hydrological, geological, morphological, and pedological characteristics and land use and occupation, a reclassification of the selected input layers was performed. Scores were first defined for each subclassification of the geospatial input layer, with variation ranging from 1 to 5. In the soil drainage layer, the scores ranged from 0.5 to 5, due to a greater number of classifications.

Each score was considered in terms of its ability to promote flooding events, such that higher scores indicated a stronger influence to the event under analysis, in view of the construction of the flood risk map for the selected areas.

## Height

Shuttle Radar Topography Mission (SRTM) satellite images that are freely available from the United States Geological Survey (USGS 2021)with a spatial resolution of 30 m were used to extract altimetric data using GIS (Table 1). The merge and clip tool available in the Quantum-GIS software (QGIS 2020) was used to obtain the mosaic and cut images, respectively. Thus, the variation in the altimetric quota for each study area was obtained. The elevation data obtained were classified into five intervals, which were defined based on their altimetric amplitudes (Table 1), to divide the amplitude into five so that the classes could be defined. Thereafter, scores ranging from 1 to 5 were assigned, in which the areas with lower altitudes received the highest score because they are the most vulnerable to receiving runoff and, therefore, to flooding.

**Table 14** Precipitation in the study areas

| Precipitation—ordinary kriging | | | |
|---|---|---|---|
| Study areas | União da Vitória, PR | Blumenau, SC | Governador Valadares, MG |
| Minimum precipitation | 221.63 | 77.16 | 152.67 |
| Maximum precipitation | 297.85 | 154.85 | 234.35 |
| Amplitude | 76.22 | 77.69 | 81.68 |

## Slope

SRTM satellite images with a spatial resolution of 30 m also occurred in the altitude layer (Table 1) to extract the slope data using GIS. Mosaics and clippings of satellite images were first created based on the study areas. The slopes were then extracted using the slope tool available in QGIS and reclassified into six classes based on the EMBRAPA

**Table 15** Rainfall stations used in ordinary kriging

| Station | Study area | Responsible | mm |
|---|---|---|---|
| 2551017 | União da Vitória, PR | ANA | 316.2 |
| 2551018 | União da Vitória, PR | ANA | 319.0 |
| 2650037 | União da Vitória, PR | ANA | 222.8 |
| 2651000 | União da Vitória, PR | ANA | 297.9 |
| 2651004 | União da Vitória, PR | ANA | 224.0 |
| 2651005 | União da Vitória, PR | ANA | 222.0 |
| 2651013 | União da Vitória, PR | ANA | 228.3 |
| 2651016 | União da Vitória, PR | ANA | 229.1 |
| 2651020 | União da Vitória, PR | ANA | 238.5 |
| 2651026 | União da Vitória, PR | ANA | 146.0 |
| 2651065 | União da Vitória, PR | ANA | 232.5 |
| 2651066 | União da Vitória, PR | ANA | 249.6 |
| 2651076 | União da Vitória, PR | CEMADEN | 208.4 |
| 2648001 | Blumenau, SC | ANA | 107.5 |
| 2648002 | Blumenau, SC | ANA | 104.8 |
| 2649001 | Blumenau, SC | ANA | 131.0 |
| 2649002 | Blumenau, SC | ANA | 109.1 |
| 2649003 | Blumenau, SC | ANA | 103.6 |
| 2649004 | Blumenau, SC | ANA | 154.9 |
| 2649005 | Blumenau, SC | ANA | 123.0 |
| 2649007 | Blumenau, SC | ANA | 99.3 |
| 2649008 | Blumenau, SC | ANA | 77.1 |
| 2649009 | Blumenau, SC | ANA | 97.3 |
| 2649010 | Blumenau, SC | ANA | 109.5 |
| 2649065 | Blumenau, SC | ANA | 76.8 |
| 2748000 | Blumenau, SC | ANA | 126.8 |
| 2749000 | Blumenau, SC | ANA | 151.6 |
| 2749016 | Blumenau, SC | ANA | 162.0 |
| 2749045 | Blumenau, SC | ANA | 149.4 |
| 1841001 | Gov. Valadares, MG | ANA | 185.5 |
| 1841011 | Gov. Valadares, MG | ANA | 231.2 |
| 1841019 | Gov. Valadares, MG | ANA | 132.7 |
| 1841020 | Gov. Valadares, MG | ANA | 234.4 |
| 1842005 | Gov. Valadares, MG | ANA | 126.9 |
| 1941018 | Gov. Valadares, MG | ANA | 110.6 |
| 1942032 | Gov. Valadares, MG | ANA | 67.1 |

classification (Table 2) using the r.reclass tool available in QGIS. Finally, grades ranging from 1 to 5 were assigned, where slopes, according to EMBRAPA classes, received grades of 5, flat relief; 4, smooth wavy relief; 3, wavy relief; 2, strong wavy relief; and 1, mountainous and steep reliefs. Subsequently, a reclassification by table was performed using QGIS, with the assigned grades.

## Drainage distance

As floods normally occur near river banks and flood areas of the floodplain, ring buffers of 125 m, 250 m, 375 m, 500 m, and more than 500 m were generated in the main river, main tributaries, and masses of water. The files used in this survey were made available by IBGE on the IBGE Maps Portal, version 2021, at a scale of 1:250000. Grades were assigned according to proximity using digital files of the buffers, ranging from 1 to 5, where grade 5 was assigned to areas within a proximity of 125 m, grade 4 was for areas within a proximity between 125 m and 250 m, etc. Finally, rasterization of the file was performed using the QGIS tool to converting the file from a vector to raster.

## Soil drainage

For the classification of soil drainage in the study areas, the mapping of soils in Brazil, made available via the web by IBGE, version 2017, scale 1:250000, was used. The vector file was cut using QGIS for the areas under analysis.

**Table 16** Reliefs of the study areas

| Relief | Score | $km^2$ | % |
|---|---|---|---|
| União da Vitória, Paraná | | | |
| Water body | 5 | 19.5 | 1.55% |
| Patamar de Ponta Grossa, Itapetininga | 2 | 69.6 | 5.53% |
| Planalto Dissecado Rio Iguaçu | 1 | 736.2 | 58.54% |
| Planalto dos Campos Gerais | 1 | 298.8 | 23.76% |
| Planicies Alúvio-coluvionares | 4 | 133.6 | 10.63% |
| Blumenau, Santa Catarina | | | |
| Water body | 5 | 9.3 | 0.74% |
| Planícies Alúvio-coluvionares | 4 | 154.1 | 12.26% |
| Serras do Leste Catarinense | 1 | 1093.2 | 87.00% |
| Governador Valadares, Minas Gerais | | | |
| Water body | 5 | 21.0 | 1.64% |
| Depressão Interplanáltica do Médio Rio Doce | 3 | 1148.5 | 89.34% |
| Planalto da Zona Metalurgica Mineira | 1 | 11.2 | 0.87% |
| Planicies e Terraços Fluviais | 4 | 2.3 | 0.18% |
| Planicies e Terraços Fluviais do Rio Doce | 4 | 77.9 | 6.06% |
| Serranias da Zona da Mata Mineira | 1 | 24.5 | 1.91% |

Soil data were classified according to permeability (drainage), based on the soil characteristics that were identified in each study area, and were thus defined according to whether they were excessively drained, somewhat excessively drained, very well drained, well drained, moderately drained, somewhat poorly drained, poorly drained, or very poorly drained. Then, grades were assigned depending on their contribution to the generation of flood events (Table 3). In this case, owing to the existence of eight classes, it was decided to assign scores ranging from 0.5 to 5, emphasizing that the pedology shapefile used for the classification of drainage contained the urban and water body classes, which received scores of 4.5 and 5.0, respectively, both in the drainage class and in the soil hydrological group. Subsequently, the vector-to-raster conversion was performed using the QGIS software.

**Table 17** Land use and occupation in the study areas

| Code | Class | Score | $km^2$ | % |
|---|---|---|---|---|
| Map Biomas 2014—União da Vitória, Paraná | | | | |
| 3 | Forest formation | 1 | 753.55 | 59.92% |
| 9 | Forestry | 1.5 | 131.73 | 10.47% |
| 15 | Pasture | 3 | 35.70 | 2.84% |
| 21 | Agriculture and grassland mosaic | 2.5 | 197.21 | 15.68% |
| 24 | Urbanized area | 4 | 23.29 | 1.85% |
| 25 | Other non-vegetable areas | 4 | 2.67 | 0.21% |
| 33 | River, lake, and ocean | 5 | 30.89 | 2.46% |
| 39 | Soybean (temporary crop) | 2 | 36.50 | 2.90% |
| 41 | Other temporary crops | 2 | 46.09 | 3.67% |
| Map Biomas 2013—Blumenau, Santa Catarina | | | | |
| 3 | Forest formation | 1 | 763.58 | 60.77% |
| 9 | Forestry | 1.5 | 11.83 | 0.94% |
| 15 | Pasture | 3 | 43.82 | 3.49% |
| 21 | Agriculture and grassland mosaic | 2.5 | 210.20 | 16.73% |
| 24 | Urbanized area | 4 | 187.87 | 14.95% |
| 25 | Other non-vegetable areas | 4 | 1.16 | 0.09% |
| 33 | River, lake, and ocean | 5 | 12.73 | 1.01% |
| 39 | Soybean (temporary crop) | 2 | 0.01 | 0.001% |
| 40 | Rice (beta)—temporary crop | 2 | 2.61 | 0.21% |
| 41 | Other temporary crops | 2 | 22.79 | 1.81% |
| Map Biomas 2013— Governador Valadares, Minas Gerais | | | | |
| 3 | Forest formation | 1 | 80.85 | 6.29% |
| 4 | Savanna formation—forest | 1 | 86.98 | 6.77% |
| 9 | Forestry | 1.5 | 0.43 | 0.03% |
| 11 | Flooded field and swamp area | 5 | 1.11 | 0.09% |
| 15 | Pasture | 3 | 966.73 | 75.20% |
| 21 | Agriculture and grassland mosaic | 2.5 | 78.15 | 6.08% |
| 24 | Urbanized area | 4 | 45.11 | 3.51% |
| 25 | Other non-vegetable areas | 4 | 1.06 | 0.08% |
| 29 | Rocky outcrop | 1 | 0.81 | 0.06% |
| 33 | River, lake, and ocean | 5 | 24.29 | 1.89% |
| 46 | Coffee (beta)—perennial crop | 2 | 0.02 | 0.002% |

**Table 18** Results from the União da Vitória, PR, study area

| Class | Area | | Classification | Weight (AHP) |
|---|---|---|---|---|
| | $km^2$ | % | | |
| Altitude (m) | | | | |
| Até 800 | 329.24 | 26.18% | 5 | 0.0173 |
| 800–900 | 373.89 | 29.73% | 4 | |
| 900–1000 | 374.59 | 29.79% | 3 | |
| 1000–1100 | 158.20 | 12.58% | 2 | |
| > 1100 | 21.71 | 1.73% | 1 | |
| Slope | | | | |
| 0–3% | 71.29 | 5.67% | 5 | 0.0311 |
| 3–8% | 242.96 | 19.32% | 4 | |
| 8–20% | 488.91 | 38.88% | 3 | |
| 20–45% | 410.97 | 32.68% | 2 | |
| 45–75% | 41.99 | 3.34% | 1 | |
| > 75% | 1.49 | 0.12% | 1 | |
| Drainage distance (m) | | | | |
| < 125 | 176.52 | 14.04% | 5 | 0.2167 |
| 125–250 | 150.23 | 11.95% | 4 | |
| 250–375 | 142.74 | 11.35% | 3 | |
| 375–500 | 132.73 | 10.55% | 2 | |
| > 500 | 655.39 | 52.11% | 1 | |
| Soil drainage | | | | |
| Overly drained | 85.28 | 6.78% | 0.5 | 0.0685 |
| Heavily drained | 744.07 | 59.16% | 1 | |
| Very well drained | 64.18 | 5.10% | 1.5 | |
| Well drained | 72.95 | 5.80% | 2 | |
| Moderately drained | 180.42 | 14.35% | 2.5 | |
| Very poorly drained | 65.20 | 5.18% | 4 | |
| Urban area | 26.05 | 2.07% | 4.5 | |
| Continental water bodies | 19.47 | 1.55% | 5 | |
| Soil hydrological group | | | | |
| Group A | - | - | 1 | 0.0642 |
| Group B | 84.2 | 6.70% | 2 | |
| Group C | 913.2 | 72.62% | 3 | |
| Group D | 214.7 | 17.07% | 4 | |
| Urban area | 26.1 | 2.07% | 5 | |
| Continental water bodies | 19.5 | 1.55% | 5 | |
| Precipitation (mm) | | | | |
| 220–237 | 449.49 | 35.74% | 1 | 0.3998 |
| 237–254 | 319.62 | 25.41% | 2 | |
| 254–271 | 209.01 | 16.62% | 3 | |
| 271–288 | 242.49 | 19.28% | 4 | |
| 288–305 | 37.03 | 2.94% | 5 | |
| Relief | | | | |
| Water body | 19.47 | 1.55% | 5 | 0.0492 |
| Patamar de Ponta Grossa—Itapetininga | 69.56 | 5.53% | 2 | |
| Planalto Dissecado Rio Iguaçu | 736.21 | 58.54% | 1 | |
| Planalto dos Campos Gerais | 298.76 | 23.76% | 1 | |
| Planicies Alúvio-coluvionares | 133.62 | 10.63% | 4 | |

Table 18 (continued)

| Class | Area | | Classification | Weight (AHP) |
|---|---|---|---|---|
| | $km^2$ | % | | |
| Land use and coverage—Map Biomas 2014 | | | | |
| Forest formation | 753.55 | 59.92% | 1 | 0.1533 |
| Forestry | 131.73 | 10.47% | 1.5 | |
| Soybean (temporary crop) | 36.50 | 2.90% | 2 | |
| Other temporary crops | 46.09 | 3.67% | 2 | |
| Agriculture and grassland mosaic | 197.21 | 15.68% | 2.5 | |
| Pasture | 35.70 | 2.84% | 3 | |
| Urbanized area | 23.29 | 1.85% | 4 | |
| Other non-vegetable areas | 2.67 | 0.21% | 4 | |
| River, lake, and ocean | 30.89 | 2.46% | 5 | |

## Soil hydrological group

Based on the infiltration rate and runoff potential, the soil texture classes were reclassified into the four main soil hydrological groups (groups A, B, C, and D) using the soil mapping shapefile of Brazil, version 2017, on a scale of 1:250000, made available by the IBGE.

Infiltration varies according to the soil texture and controls surface runoff and flooding. Based on the infiltration rate and runoff potential, the soil classes were reclassified into soil groups to be assigned grades, depending on their contribution to flooding events, as shown in Table 4.

## Precipitation

For the collection of rainfall data, data from the National Agency for Water and Basic Sanitation (ANA 2022) and the National Center for Monitoring and Alerts for Natural Disasters (CEMADEN 2021) were used, as they were publicly available data within close proximity to the study area (Fig. 2).

Such data were used for the elaboration of the precipitation map from the ordinary kriging tool of geostatistics available through the installation of the computational application R (R-PROJECT 2020) in QGIS, in which the output data had a spatial resolution of 30 m.

The precipitation data used refer to three consecutive days of rain, which were identified in FIDE, which describes the flood event in the selected study areas.

## Relief

To identify the landforms in the three study areas, the shapefile file made available by the Brazilian Institute of Geography and Statistics (IBGE 2022) was used at a scale of 1:250000, identified using Geomorphology version 2017, and obtained via the web at the IBGE map portal site.

Clipping of the digital files was performed using GIS for the areas under analysis, and grades ranging from 1 to 5 were assigned to each existing class according to their contribution to generating flood events. Finally, the transformation from vector to raster was performed using QGIS software.

## Land use and occupation

The use and cover of soil influence the infiltration, evapotranspiration, and generation of surface runoff and were, therefore, of paramount importance in this study. To obtain the use and occupation of land in the three study areas, information provided by Map Biomas from collection 6, published in August 2021 was used, which refers to the year of the flood event, as presented in FIDE. Thereafter, the clip raster tool in the QGIS software was used to identify the classes of land use and occupation and, finally, to reclassify them according to the scores adopted in each class, ranging from 1 to 5, in which they represent collaboration in generating flood events.

## Analytical hierarchical analysis

For each geospatial dataset used in the characterization of flood risk areas, a multicriteria analysis was applied using the AHP to establish priorities, which indicate the input layers with the greatest influence on causing floods.

Altitude, slope, precipitation, soil drainage, soil hydrological group, drainage distance, relief, and land use and cover were defined by the multicriteria method using the AHP.

The AHP, proposed by Saaty (1977), is a multi-objective decision analysis with several criteria, according to which importance weights are assigned, and a pairwise comparison is outperformed to establish the degree of importance of each criterion.

The paired matrix was normalized so that the elements of each column were summed; therefore, each element was divided by the sum of its columns. To identify the weight vector, the average of each row of the normalized matrix was calculated.

The consistency index indicates the coherence in the judgments of priorities assigned in the matrix to certify its consistency; thus, the consistency index was calculated from Eq. (1):

**Table 19** Results of the study area of Blumenau, SC

| Class | Area | | Classification | Peso (AHP) |
|---|---|---|---|---|
| | $km^2$ | % | | |
| Altitude (m) | | | | |
| < 200 | 765.73 | 60.94% | 5 | 0.0173 |
| 200–400 | 380.73 | 30.30% | 4 | |
| 400–600 | 99.36 | 7.91% | 3 | |
| 600–800 | 10.55 | 0.84% | 2 | |
| > 800 | 0.22 | 0.02% | 1 | |
| Slope | | | | |
| 0–3% | 54.67 | 4.35% | 5 | 0.0311 |
| 3–8% | 177.85 | 14.15% | 4 | |
| 8–20% | 315.76 | 25.13% | 3 | |
| 20–45% | 563.35 | 44.83% | 2 | |
| 45–75% | 139.31 | 11.09% | 1 | |
| > 75% | 5.65 | 0.45% | 1 | |
| Drainage distance (m) | | | | |
| < 125 | 246.87 | 19.65% | 5 | 0.2167 |
| 125–250 | 219.26 | 17.45% | 4 | |
| 250–375 | 197.56 | 15.72% | 3 | |
| 375–500 | 167.74 | 13.35% | 2 | |
| > 500 | 425.15 | 33.83% | 1 | |
| Soil drainage | | | | |
| Heavily drained | 705.59 | 56.15% | 1 | 0.0685 |
| Moderately drained | 385.08 | 30.65% | 2.5 | |
| Very poorly drained | 18.28 | 1.45% | 4 | |
| Urban area | 138.38 | 11.01% | 4.5 | |
| Continental water bodies | 9.26 | 0.74% | 5 | |
| Soil hydrological group | | | | |
| Group A | - | - | 1 | 0.0642 |
| Group B | - | - | 2 | |
| Group C | 1090.68 | 86.80% | 3 | |
| Group D | 18.28 | 1.45% | 4 | |
| Urban area | 138.38 | 11.01% | 5 | |
| Continental water bodies | 9.26 | 0.74% | 5 | |
| Precipitation (mm) | | | | |
| 77–94 | 24.44 | 1.95% | 1 | 0.3998 |
| 94–111 | 232.52 | 18.50% | 2 | |
| 111–128 | 932.85 | 74.24% | 3 | |
| 128–145 | 59.00 | 4.70% | 4 | |
| 145–162 | 7.77 | 0.62% | 5 | |
| Relief | | | | |
| Water body | 9.26 | 0.74% | 5 | 0.0492 |
| Planícies Alúvio-coluvionares | 154.11 | 12.26% | 4 | |
| Serras do Leste Catarinense | 1093.22 | 87.00% | 1 | |

Table 19 (continued)

| Class | Area | | Classification | Peso (AHP) |
|---|---|---|---|---|
| | km² | % | | |
| Use and occupation of land | | | | |
| Forest formation | 763.58 | 60.77% | 1 | 0.1533 |
| Forestry | 11.83 | 0.94% | 1.5 | |
| Soybean (temporary crop) | 0.01 | 0.00% | 2 | |
| Rice (beta)—temporary crop | 2.61 | 0.21% | 2 | |
| Other temporary crops | 22.79 | 1.81% | 2 | |
| Agriculture and grassland mosaic | 210.20 | 16.73% | 2.5 | |
| Pasture | 43.82 | 3.49% | 3 | |
| Urbanized area | 187.87 | 14.95% | 4 | |
| Other non-vegetable areas | 1.16 | 0.09% | 4 | |
| River, lake, and ocean | 12.73 | 1.01% | 5 | |

$$CR = \frac{CI}{RI} \quad (1)$$

where CI is the consistency index and RI is the random inconsistency index. If CR ≤ 0.1 or CR ≤ 10%, the matrix is considered to be consistent, and at a value greater than 10%, the comparisons are reviewed.

The maximum eigenvalue (λmáx) was calculated using Eq. (2).

$$\lambda \max = \frac{1}{n} \sum_{i=1}^{n} \frac{AWi}{Wi} \quad (2)$$

The random consistency index directly depends on the number of conditioning factors used in the paired matrix. The consistency index (CI) was calculated using Eq. (3).

$$CI = \frac{\lambda \max - n}{(n-1)} \quad (3)$$

where $n$ is equivalent to the number of criteria adopted in the comparison matrix.

Saaty (1991) proposed a table with random indices (RI), as listed in Table 5.

After assigning the weights of the parameters considered, with their subclassifications and the established scores, in possession of the database in raster format, this information was processed using GIS, specifically the raster calculator tool, which was used to obtain the weighted sum of the data from the input, as a result of this layer, as well as the flood risk map.

The modeled output was reclassified into five vulnerability classes: very low, low, moderate, high, and very high, using a table reclassification tool available in QGIS. Thus, the final representation of the map was aimed at areas at risk of flooding. Subsequently, areas most susceptible to flooding were identified, making it possible to analyze the similarities and differences found in the three areas, which are located in the same Brazilian biome, but in different cities.

### Flood risk map

The flowchart in Fig. 3 describes the methodological structure of this study.

## Results and discussion

### Analytical hierarchical analysis

The paired matrix was filled, where the preference of one criterion was compared over another. Specifically, the criterion in the row was compared with that in the column, and comparisons were made by determining how much more important the element to the left of the matrix was perceived in relation to the analyzed property in question (floods), compared to the element at the top of the array. For peer comparison judgments, a scale of 1 to 9 was used, as shown in Table 6.

The row criterion was compared with the column to fill in the paired matrix; the priorities are listed in Table 7.

Therefore, as a result of the AHP, the following priorities and respective weights were obtained according to Table 8, which were used in the map algebra.

For the elaborated comparison matrix, a consistency index (CR) equivalent of 0.213 was obtained. However, as a rule, a value lower than or equal to 0.1 is required (Saaty 1991), but the consistency found to be valid is justified due to the problem addressed considering different parameters, since the input classes cover eight different pieces of information.

**Table 20** Results of the study area of Governador Valadares, MG

| Class | Area | | Classification | Weight (AHP) |
|---|---|---|---|---|
| | $km^2$ | % | | |
| Altitude (m) | | | | |
| < 300 | 1068.89 | 83.15% | 5 | 0.0173 |
| 300–500 | 194.47 | 15.13% | 4 | |
| 500–700 | 12.82 | 1.00% | 3 | |
| 700–900 | 7.72 | 0.60% | 2 | |
| > 900 | 1.64 | 0.13% | 1 | |
| Slope | | | | |
| 0–3% | 84.38 | 6.56% | 5 | 0.0311 |
| 3–8% | 172.20 | 13.39% | 4 | |
| 8–20% | 349.47 | 27.18% | 3 | |
| 20–45% | 610.25 | 47.47% | 2 | |
| 45–75% | 66.59 | 5.18% | 1 | |
| > 75% | 2.65 | 0.21% | 1 | |
| Drainage distance (m) | | | | |
| < 125 | 288.47 | 22.44% | 5 | 0.2167 |
| 125–250 | 245.69 | 19.11% | 4 | |
| 250–375 | 213.76 | 16.63% | 3 | |
| 375–500 | 171.72 | 13.36% | 2 | |
| > 500 | 365.90 | 28.46% | 1 | |
| Soil drainage | | | | |
| Very well drained | 54.36 | 4.23% | 1.5 | 0.0685 |
| Well drained | 1143.98 | 88.99% | 2 | |
| Imperfectly drained | 24.95 | 1.94% | 3 | |
| Poorly drained | 4.38 | 0.34% | 3.5 | |
| Urban area | 38.56 | 3.00% | 4.5 | |
| Continental water bodies | 19.30 | 1.50% | 5 | |
| Soil hydrological group | | | | |
| Group A | 146.57 | 11.40% | 1 | 0.0642 |
| Group B | 838.76 | 65.25% | 2 | |
| Group C | 196.07 | 15.25% | 3 | |
| Group D | 46.28 | 3.60% | 4 | |
| Urban area | 38.56 | 3.00% | 5 | |
| Continental water bodies | 19.30 | 1.50% | 5 | |
| Precipitation (mm) | | | | |
| 152–169 | 1075.82 | 83.69% | 1 | 0.3998 |
| 169–186 | 112.09 | 8.72% | 2 | |
| 186–203 | 59.78 | 4.65% | 3 | |
| 203–220 | 30.21 | 2.35% | 4 | |
| 220–237 | 7.64 | 0.59% | 5 | |
| Relief | | | | |
| Water body | 21.04 | 1.64% | 5 | 0.0492 |
| Depressão Interplanáltica do Médio Rio Doce | 1148.51 | 89.34% | 3 | |
| Planalto da Zona Metalurgica Mineira | 11.21 | 0.87% | 1 | |
| Planicies e Terraços Fluviais | 2.33 | 0.18% | 4 | |
| Planicies e Terraços Fluviais do Rio Doce | 77.90 | 6.06% | 4 | |
| Serranias da Zona da Mata Mineira | 24.54 | 1.91% | 1 | |

Table 20 (continued)

| Class | Area | | Classification | Weight (AHP) |
|---|---|---|---|---|
| | $km^2$ | % | | |
| Land use and coverage | | | | |
| Forest formation | 80.85 | 6.29% | 1 | 0.1533 |
| Savanna formation—forest | 86.98 | 6.77% | 1 | |
| Rocky outcrop | 0.81 | 0.06% | 1 | |
| Forestry | 0.43 | 0.03% | 1.5 | |
| Coffee (beta)—perennial crop | 0.02 | 0.002% | 2 | |
| Agriculture and grassland mosaic | 78.15 | 6.08% | 2.5 | |
| Pasture | 966.73 | 75.20% | 3 | |
| Urbanized area | 45.11 | 3.51% | 4 | |
| Other non-vegetable areas | 1.06 | 0.08% | 4 | |
| Flooded field and swamp area | 1.11 | 0.09% | 5 | |
| River, lake, and ocean | 24.29 | 1.89% | 5 | |

## Altitude

The altitudes differed in the three areas; however, a standard behavior can be seen, where the largest portion of each area is found at lower altitudes (Table 9).

## Slope

The slopes obtained according to the EMBRAPA classification demonstrate that the class of wavy relief predominated in the study area of União da Vitória, PR; in Blumenau,SC, and Governador Valadares, the predominant reliefs were the strong wavy relief (Table 10).

## Drainage distance

The drainage distances found for the areas under analysis are presented in Table 11, which shows that the areas closest to the drainage do not represent the largest shares in the study areas. However, it is known that the urban area is located on the banks of the main river, which is an important factor in the analysis of drainage proximity.

## Soil drainage

Scores were assigned according to those in Table 3, and the results are shown in Table 12. According to the existing soils, efficient drainage generally occurred in most of the areas.

## Soil hydrological group

The classification of hydrological groups in the three study areas differed, and there were the classes of B, C, and D in União da Vitória, PR; C and D in Blumenau, SC; and A, B, C, and D in Governador Valadares, MG (Table 13).

## Precipitation

The following results were obtained using ordinary kriging, as shown in Table 14, in which the differences in rainfall in the three study areas were verified.

The rainy dates used in União da Vitória, PR, were June 6, 7, and 8, 2014; in Blumenau, SC, they were September 20, 21, and 22, 2013; in Governadores Valadares, MG, they were December 17, 18, and 19, 2013. The precipitation points used are described in Table 15.

## Relief

The relief classifications and assigned scores are listed in Table 16, which shows that each area had a different type of relief.

## Land use and occupation

The land use and occupation classes are listed in Table 17.

The predominance of forest formation was observed in the study areas of União da Vitória, PR, and Blumenau, SC, with more than half of their areas being covered by forests, while in Governador Valadares, MG, approximately 75% represented areas occupied by pastures.

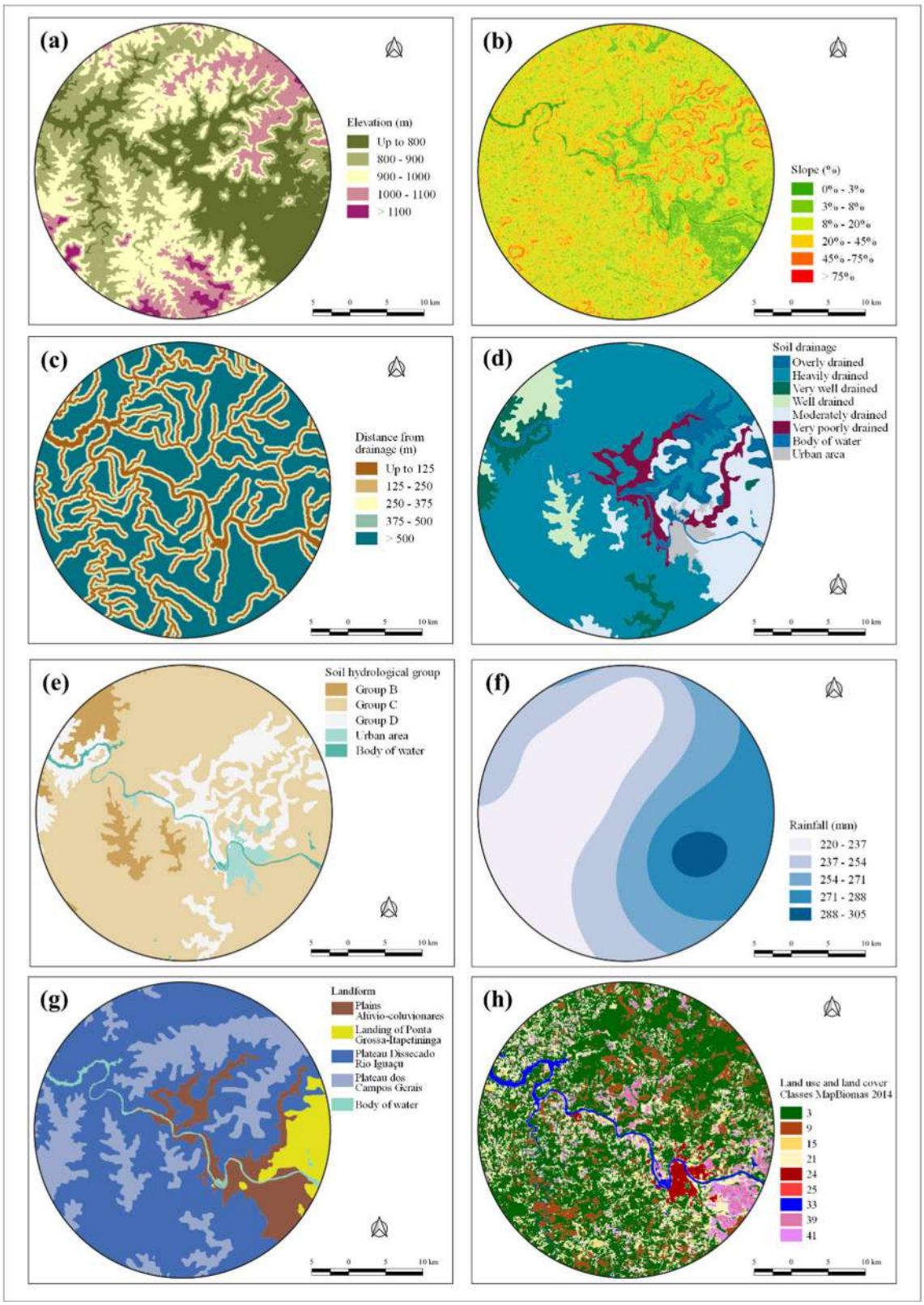



**Fig. 4** Conditioning parameters used for flood-hazard susceptibility mapping (União da Vitória, PR). (**a**) elevation, (**b**) slope, (**c**) distance from drainage, (**d**) soil drainage, (**e**) hydrological soil group, (**f**) rainfall, (**g**) landform, and (**h**) land use and land cover

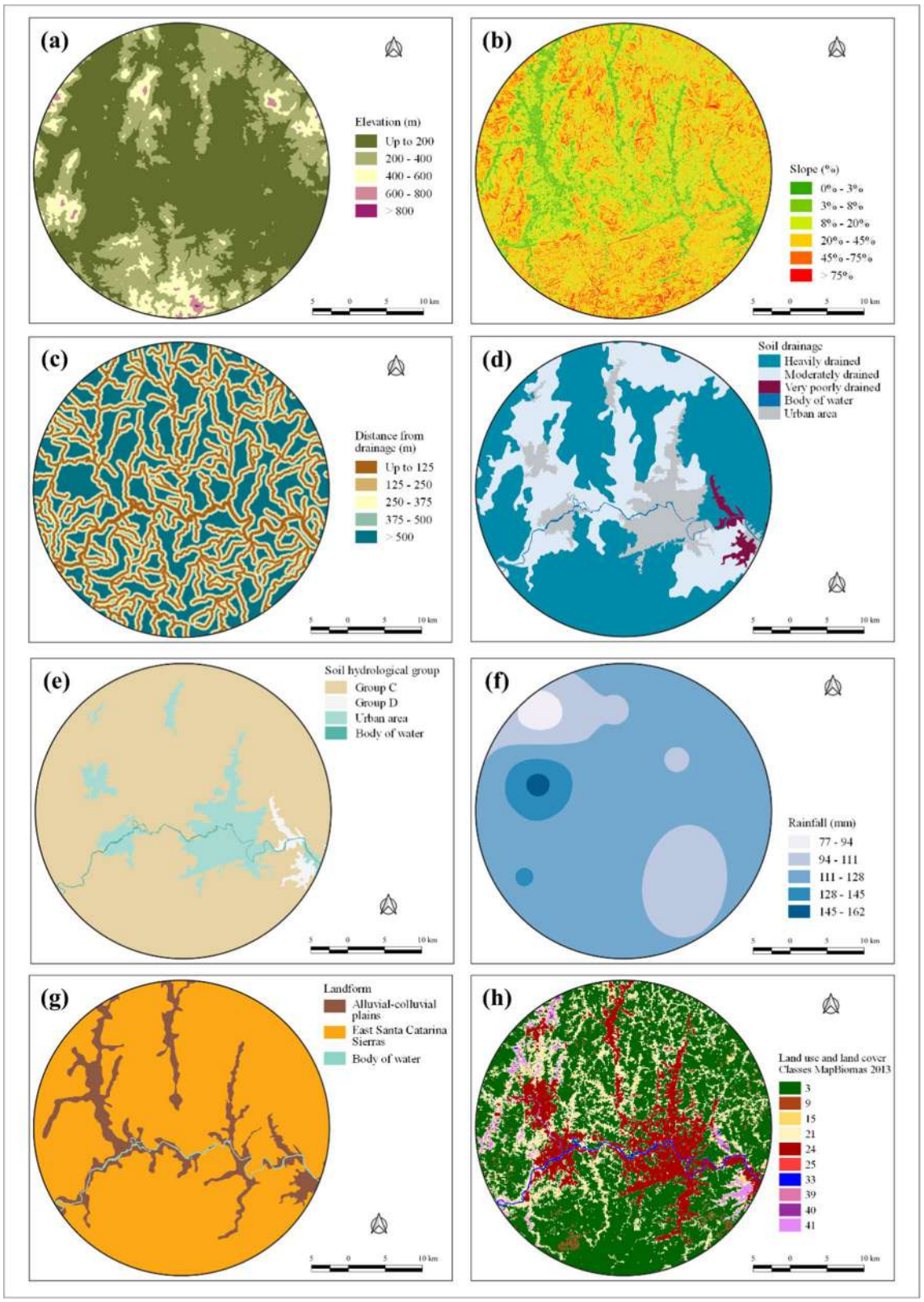



**Fig. 5** Conditioning parameters used for flood-hazard susceptibility mapping (Blumenau, SC). (**a**) elevation, (**b**) slope, (**c**) distance from drainage, (**d**) soil drainage, (**e**) hydrological soil group, (**f**) rainfall, (**g**) landform, and (**h**) land use and land cover

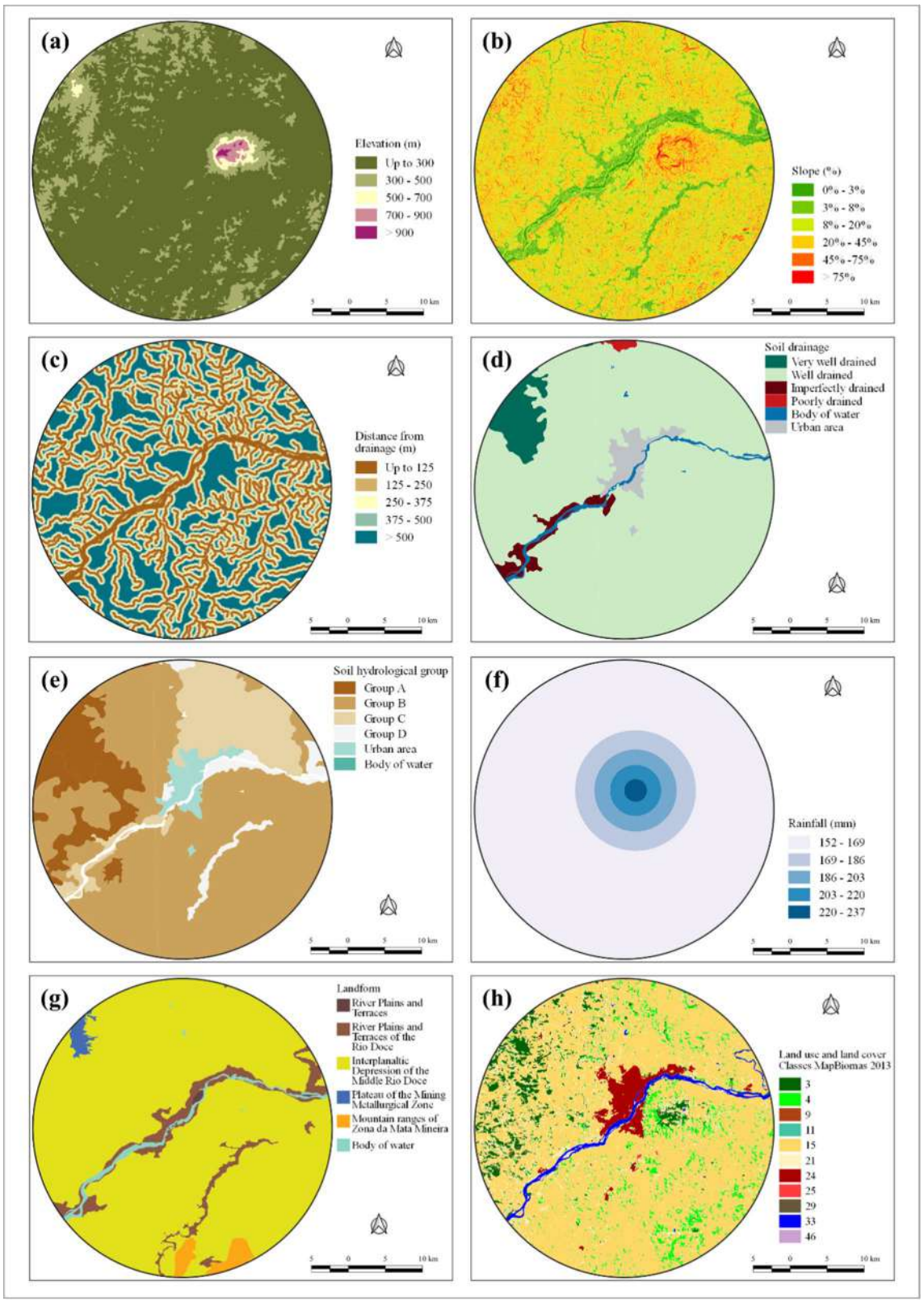



**Fig. 6** Conditioning parameters used for flood-hazard susceptibility mapping (Governador Valadares, MG). (**a**) elevation, (**b**) slope, (**c**) distance from drainage, (**d**) soil drainage, (**e**) hydrological soil group, (**f**) rainfall, (**g**) landform, and (**h**) land use and land cover

**Table 21** Flood risk areas in the study areas

| Flood risk zones | Flood risk index | Area km$^2$ | % |
|---|---|---|---|
| Study area—União da Vitória, PR | | | |
| Very low | 1.0–1.8 | 427.11 | 33.96% |
| Short | 1.8–2.6 | 518.07 | 41.19% |
| Moderate | 2.6–3.4 | 223.65 | 17.78% |
| High | 3.4–4.2 | 75.34 | 5.99% |
| Very high | 4.2–5.0 | 13.46 | 1.07% |
| Study area—Blumenau, SC | | | |
| Very low | 1.0–1.8 | 66.30 | 5.28% |
| Short | 1.8–2.6 | 621.03 | 49.42% |
| Moderate | 2.6–3.4 | 457.16 | 36.38% |
| High | 3.4–4.2 | 105.54 | 8.40% |
| Very high | 4.2–5.0 | 6.57 | 0.52% |
| Study area—Governador Valadares, MG | | | |
| Very low | 1.0–1.8 | 350.55 | 27.27% |
| Short | 1.8–2.6 | 761.30 | 59.22% |
| Moderate | 2.6–3.4 | 132.60 | 10.32% |
| High | 3.4–4.2 | 31.22 | 2.43% |
| Very high | 4.2–5.0 | 9.86 | 0.77% |

## Summary of results

A general summary of the results is presented in Tables 18, 19, and 20.

The input maps are shown in Figs. 4, 5, and 6. In these maps, exclusively in the land use and occupation layers, the classes were labeled using the codes from MapBiomas (Table 17).

## Flood risk maps

The risk areas were reclassified into five classes according to the amplitude determined, and the results are presented in Table 21.

The maps obtained from map algebra are shown in Figs. 7, 8, and 9.

The influence of the precipitation map on the development of the flood risk map is visible, since it is a decisive layer, but it is clear that for a more accurate representation of reality, a greater number of precipitation stations is necessary.

## Conclusion

Considering the analyses performed, from the multicriteria approach in line with the application of GIS, the main conclusions obtained are as follows. Since the proposed scientific methodology was conducted in three selected study areas, using data on altitude, slope, precipitation, soil drainage, soil hydrological group, drainage distance, relief and land use, and cover, comparisons and analyses of existing patterns and distinctions were made possible.

For greater accuracy of precipitation mapping and its higher intensity ranges, a greater number of pluviometric stations is necessary. However, from the data used, it was noted that the highest ranges of precipitation intensity covered an area of less than 3% in the analyzed areas, which indicates that the selected cities do not receive the largest amount of rain; however, the rain catchment area directly influences the downstream flow, which contributes to the elevation of the channel level and, consequently, the flooding of the banks of this water body.

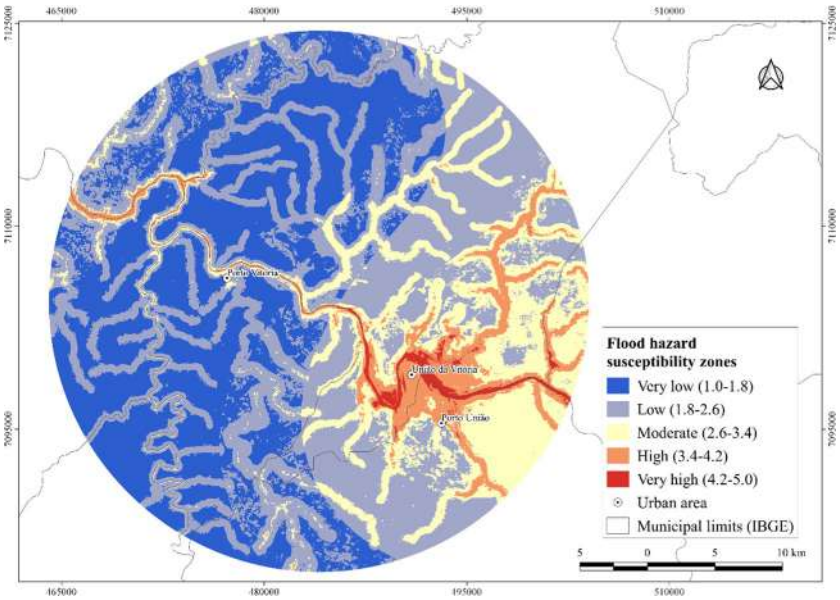



**Fig. 7** Flood-risk susceptibility map (União da Vitória, PR)

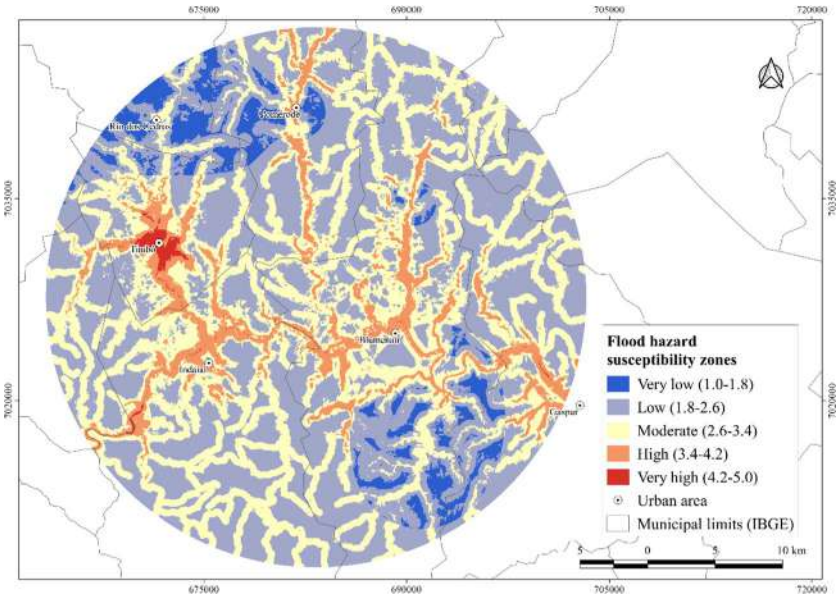



**Fig. 8** Flood-risk susceptibility map (Blumenau, SC)

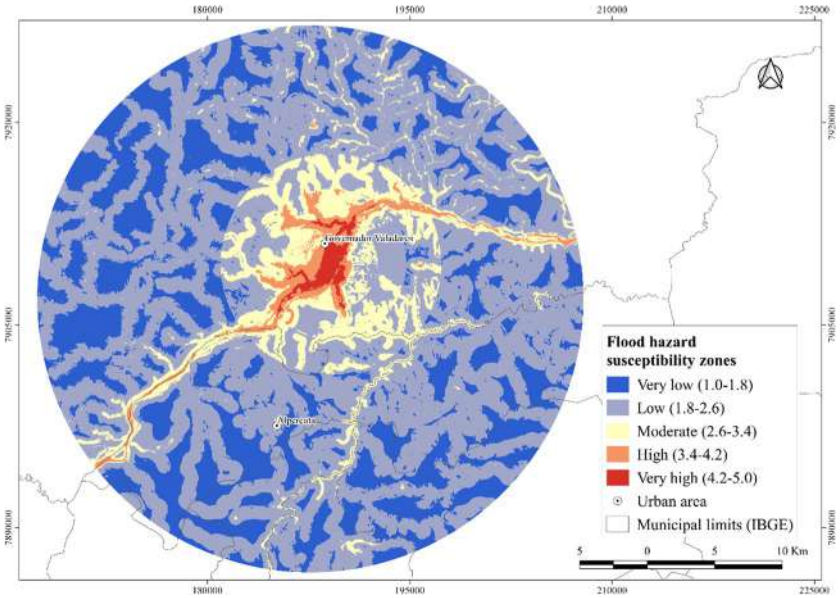



**Fig. 9** Flood-risk susceptibility map (Governador Valadares, MG)

It is a fact that consecutive days with high-intensity rain are recurring patterns in flood events in such a way that the drainage channel does not support the flow received and overflows, causing river floods, which was observed in the three study areas.

Urbanized areas did not cover most of the study areas, but they were located in very high-risk and high-risk zones on the banks of the main channel. In addition to promoting the waterproofing of the soil, reducing the infiltration of rainwater, and increasing runoff, because they are also closer to the drainage channels, when the channel overflows, they will certainly be the first areas affected.

Therefore, the lower altitudes are more susceptible to receiving runoff and cover most of the three study areas.

From the analyses performed, both the input data and the results show that the decisive and identified criteria in the three case studies were the layers of altitude and proximity to river channels.

Therefore, regions close to the main body of water, which are located at constant altitudes, with little variation of the altimetric quota on the banks of water courses, will suffer from flood disasters when there is an incidence of high-magnitude rain, in which the volume of rain expected in one month, or a few months, occurs on consecutive days.

Through the geospatial data made available by the IBGE and used in this study, the water mass vector layer had features in the three areas. Therefore, it is possible to observe significant main rivers, as their water masses were mapped using GIS, which was commonly used for the studied areas.

The Atlantic Forest biome completely covers the three study areas, and according to the analysis of land use and occupation from the Map Biomas, the three study areas had different levels of conservation, because the study areas of União da Vitória, PR, and Blumenau, SC, had approximately 60% of forest formation Governador Valadares, MG, had approximately 7% of forest and savanna formation. In addition, it is possible to observe that the vegetation did not directly influence the amount of area classified as high risk and very high risk, since in the areas of União da Vitória, PR; Blumenau, SC; and Governador Valadares, MG, approximately 7%, 9%, and 3% were high-risk and very high-risk areas, respectively.

All the layers of information considered in the adopted methodology have a certain degree of influence on the vulnerability of flood risk, but patterns that were found, and that are visible in the maps, are those of altitude, proximity to channels, and urban areas located on the banks of the water body.

Flood risk maps provide key information for mitigating the consequences of flood disasters in affected areas. Floods are natural phenomena that cannot be completely avoided, but it is possible to minimize the damage caused, mainly by prioritizing actions and urban infrastructure works in the highest-risk areas.

Public authorities are responsible for preventing and mitigating the damage caused by disasters, and flood risk mapping will help provide subsidies for mitigating actions.

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